

# SimpleRNN - Training Results

## Summary

A SimpleRNN was trained to classify text into 5 toxic categories using toxic\_data.csv. Each input is the query joined with its image description. Duplicate rows were removed before splitting the data into 70% train, 15% validation and 15% test, and 4 classes with only 2-5 unique examples were excluded. The model was trained with class weights and early stopping. On the test set it reached a macro F1 of **0.784** and a weighted F1 of **0.888**, meeting the  $F1 \geq 0.75$  requirement. Most errors are between Safe and Unknown S-Type, which share the same image descriptions.

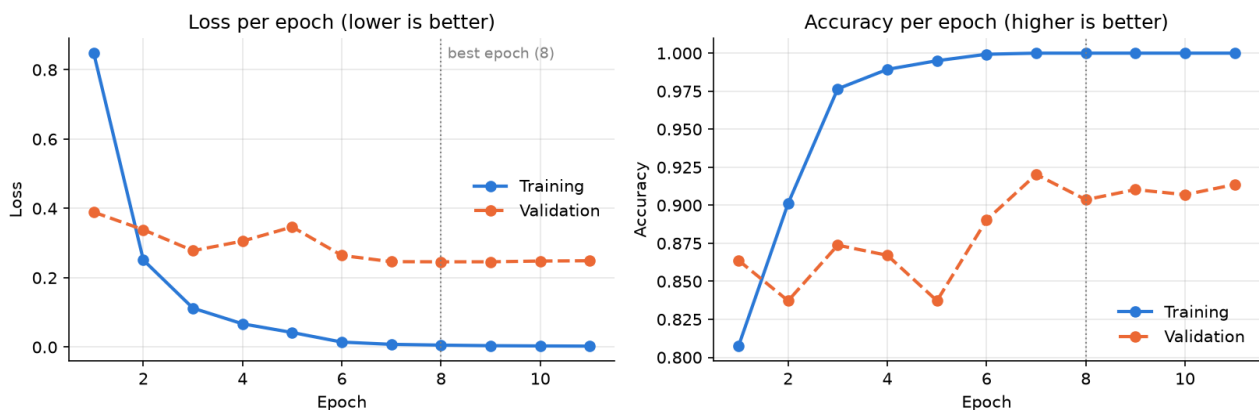
## Test results

|           | Macro | Weighted |
|-----------|-------|----------|
| F1        | 0.784 | 0.888    |
| Precision | 0.790 | 0.913    |
| Recall    | 0.790 | 0.868    |

| Class              | Precision | Recall | F1    | Test rows |
|--------------------|-----------|--------|-------|-----------|
| Non-Violent Crimes | 1.000     | 0.935  | 0.967 | 31        |
| Safe               | 0.905     | 0.795  | 0.847 | 132       |
| Unknown S-Type     | 0.100     | 0.231  | 0.140 | 13        |
| Violent Crimes     | 0.990     | 0.990  | 0.990 | 104       |
| unsafe             | 0.957     | 1.000  | 0.978 | 22        |

## Training and validation curves

SimpleRNN training history



# Confusion matrix (test set)

